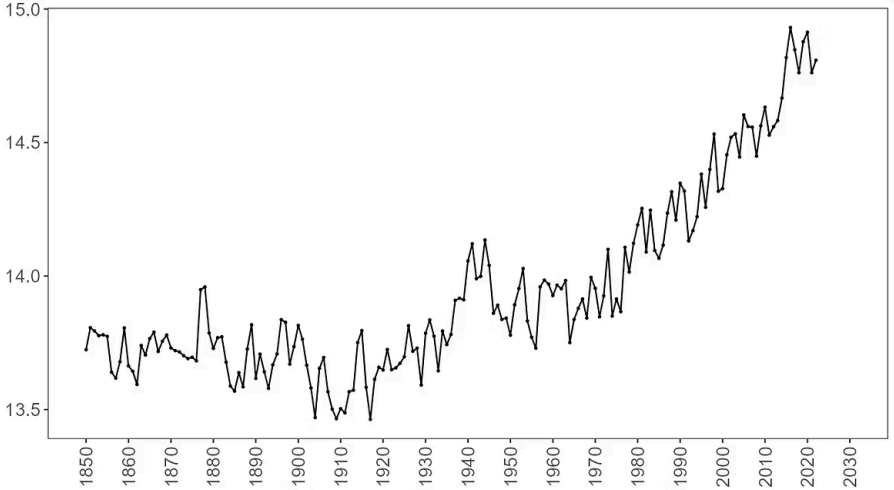
Time Series Event Detection: Anomalies, Change Points, Motifs

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What is an event in a time series?

An event is an instant or interval in which the series exhibits a **relevant change** — a rupture, anomaly, recurring pattern, or regime transition.

Relevance is always relative to the domain: what constitutes an event in climate may not carry the same meaning in finance, health, or security.

The goal of event detection is to **automatically identify** these changes and extract useful information from them for analysis or decision-making.

YGT obtained from NOAA: a univariate series used as a recurring example.

Basic formalization of time series

A time series X is an ordered sequence of n observations.

Each observation xt is indexed by time t.

The order of the data is part of the meaning: permuting the sequence destroys temporal information.

$$X = \langle x1, x2, x3, \ldots, xn \rangle,\ |X| = n,\ t \in \{1, \ldots, n\}$$

Taxonomy of Events in Time Series

- **Types of point events** — anomalies, change points, motifs, and discords
- **Data structure** — temporal granularity and dimensionality
- **Detection scenarios** — offline, online, and prediction
- **Method families** — regression, classification, clustering, and statistical
- **Evaluation** — performance metrics and computational cost

Point events: general idea

A point event occurs when a temporal characteristic deviates from what is expected.

The expectation can be estimated from the past, the future, or the divergence between both sides.

This view allows anomalies and change points to be unified.

tc(xt): value, trend, or volatility at t

Point Events: Formalization

ep(tc(xt), k): expected characteristic based on the k previous observations.

ef(tc(xt), k): expected characteristic based on the k following observations.

σ defines the tolerance for considering a change significant.

$$e(X,k,\sigma) = \{t : |tc(xt)-ep(tc(xt),k)|>\sigma \vee |tc(xt)-ef(tc(xt),k)|>\sigma \vee |ep(tc(xt),k)-ef(tc(xt),k)|>\sigma\}$$

Anomalies: intuition

An anomaly is an observation that deviates from the expected behavior of the series — whether due to an extreme value, unusual pattern, or contextual deviation.

It may indicate a measurement error, equipment failure, critical incident, or external disruption.

In applied analysis, what matters is not the absolute value, but the deviation relative to the temporal context: the same temperature may be normal in July and anomalous in December.

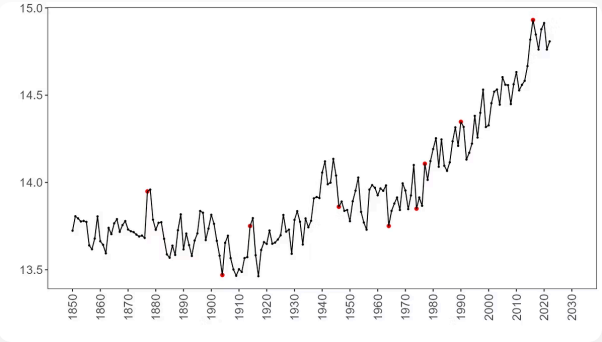
Anomalies: formalization

An observation is anomalous when it departs from what is expected both before and after instant t.

This condition separates a local irregularity from a regime transition.

$$a(X,k,\sigma) = \{t : |tc(xt)-ep(tc(xt),k)|>\sigma \wedge |tc(xt)-ef(tc(xt),k)|>\sigma\}$$

Anomalies by Residuals



Residual as an Event Signal

A model α is fitted to estimate the expected series.

The residual measures what the model did not explain.

If the model is well-fitted, extreme residuals indicate event candidates.

$$\omega t = xt - \hat{x}t, \text{ with } \hat{x}t = \alpha(xt)$$

Threshold for Extreme Residuals

Parametric: assumes approximately Gaussian residuals and uses mean ± 3 standard deviations.

Non-parametric: avoids distribution assumptions and uses quartiles and interquartile range.

$$\text{Gaussian: } \omega t \notin [\, \bar{W} - 3 \cdot sd(W),\ \bar{W} + 3 \cdot sd(W) \,]$$

$$\text{Boxplot: } \omega t \notin [\, Q1(W) - 1.5 \cdot IQR(W),\ Q3(W) + 1.5 \cdot IQR(W) \,]$$

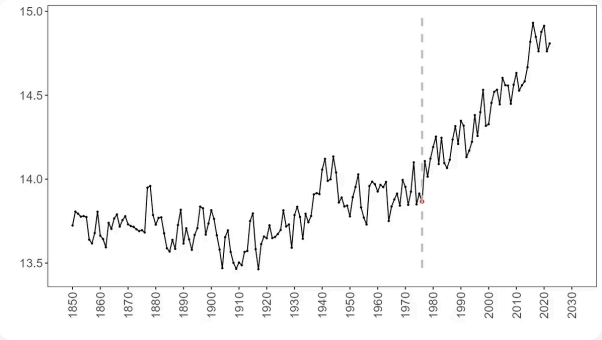
Change points: intuition

A **change point** marks a transition between distinct regimes of the series — the behavior before and after the point is structurally different.

The change can occur in the mean, variance, correlation, trend, or in a combination of parameters.

Unlike an anomaly, which is local and transitory, a change point **inaugurates a new state that persists** over time.

This distinction is fundamental for choosing the appropriate detection method.



Change points: formalization

The instant t is compatible with one side of the neighborhood, but not with both.

There is also a change when the expectations before and after t differ by more than σ.

$$cp(X,k,\sigma) = \{t : (|tc(xt)-ep(tc(xt),k)|>\sigma \square |tc(xt)-ef(tc(xt),k)|>\sigma) \vee |ep(tc(xt),k)-ef(tc(xt),k)|>\sigma\}$$

Change point as a hypothesis test

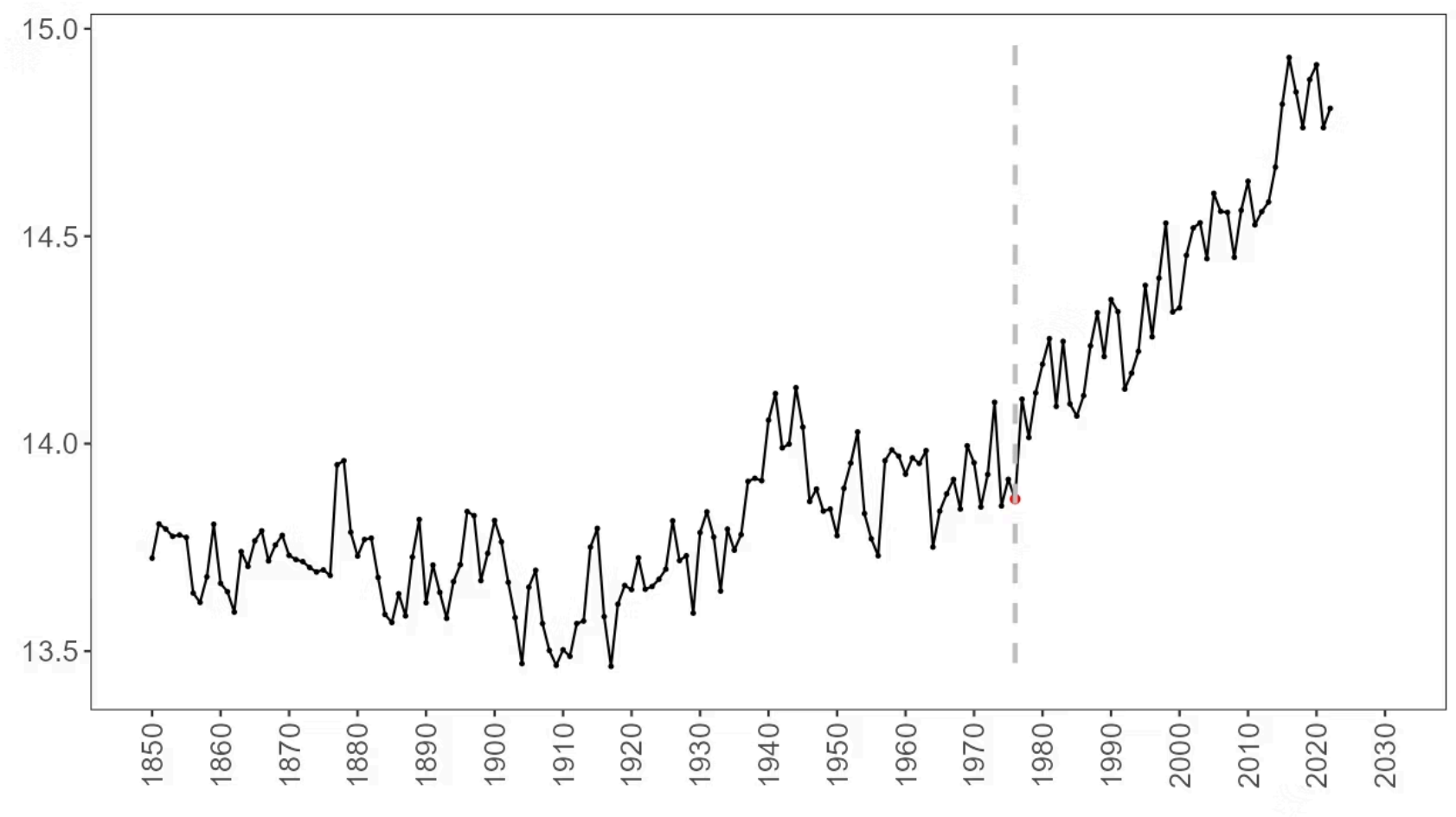
The absence of change suggests similar distributions over time.

The presence of change suggests at least one neighborhood with a different distribution.

This reading connects change point detection and concept drift.

> $H_0: P_n(t) \approx P_n(k)$ H_A: there exists k such that $P_n(t) \neq P_n(k)$

Change points in the YGT example



Dashed lines indicate change points detected with the Chow statistic.

Motifs and Discords

Motifs: repeating patterns

A **motif** is a subsequence that repeats approximately at different positions in the time series.

Recurrence reveals **latent structure**: local seasonality, process signatures, or cyclical behavior not captured by global models.

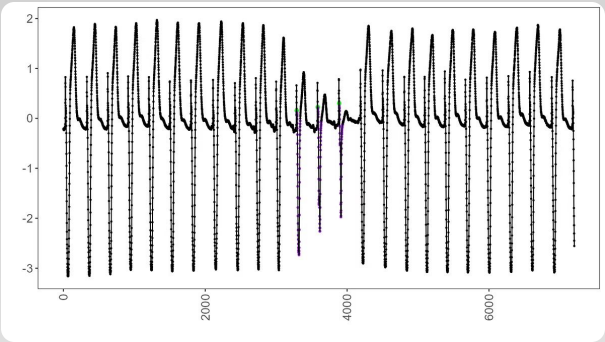
Motifs are useful for **unsupervised classification**, compression, pattern explanation, and discovery of regularities in domains such as ECG, industrial sensors, and climate series.

Discords: patterns that do not repeat

A discord is a subsequence that is very different from all others.

It can be seen as an **interval-level anomaly**.

The challenge lies in comparing subsequences without exploding the computational cost.



Motif: operational definition

Given a sequence q and a series X, q is a motif if it occurs in X at least σ times.

Discovery requires a notion of distance or similarity between subsequences.

The naive search compares many subsequences and tends to have $O(n^2)$ cost.

> q is a motif in X ⇔ occurrences(q, X) ≥ σ

Data Structure

Time series can be vectors, tables, matrices, or specialized objects.

Events can be represented by labels, intervals, timestamps, or scores.

The chosen structure directly influences the method and evaluation.

Event Granularity

Point Events

Associated with a specific instant.

Represented by a timestamp and event attributes.

Example: temperature peak in a given year.

Interval Events

Have a start and an end.

Require analysis of duration and overlap.

Example: prolonged period of intense rainfall.

year\month	1	2	3	4	5	6	7	8	9	10	11	12
1971	13.9	13.8	13.8	13.8	13.9	13.8	13.9	13.9	13.9	13.8	13.9	13.9
1972	13.7	13.7	14.0	14.0	13.9	14.0	14.0	14.0	13.9	14.0	14.0	14.0
1973	14.3	14.3	14.3	14.2	14.1	14.2	14.1	14.0	14.0	14.0	13.9	14.0
1974	13.8	13.7	13.9	13.9	13.9	13.8	13.9	14.0	13.9	13.8	13.8	13.8
1975	14.0	14.0	14.0	14.0	14.0	14.0	13.9	13.9	13.9	13.8	13.7	13.8
1976	13.9	13.8	13.8	13.9	13.8	13.9	13.9	13.9	13.9	13.7	13.9	14.0
1977	14.1	14.1	14.2	14.2	14.2	14.2	14.1	14.1	14.1	14.0	14.1	14.0
1978	14.1	14.1	14.1	14.0	14.0	14.0	14.0	13.9	14.0	14.0	14.1	14.0
1979	14.0	13.8	14.1	14.0	14.1	14.1	14.1	14.2	14.2	14.2	14.2	14.4
1980	14.3	14.3	14.2	14.2	14.3	14.2	14.2	14.1	14.1	14.1	14.2	14.1

(a)

year	value
1971	13.8
1972	13.9
1973	14.1
1974	13.8
1975	13.9
1976	13.9
1977	14.1
1978	14.0
1979	14.1
1980	14.2

(b)

Monthly global temperature aggregated to annual global temperature between 1971 and 1980.

Dimensionality

Univariate

One variable observed over time.

Simple structure, often a vector with a temporal index.

Good for introducing basic concepts and methods.

Multivariate

Two or more variables observed over time.

Allows studying relationships, dependencies, and probable causes.

Increases modeling and interpretation complexity.

value
13.8
13.9
14.1
13.8
13.9
13.9
14.1
14.0
14.1
14.2

(a)

time	value
1971	13.8
1972	13.9
1973	14.1
1974	13.8
1975	13.9
1976	13.9
1977	14.1
1978	14.0
1979	14.1
1980	14.2

(b)

time	global temperature	crude oil production
1971	13.8	2491
1972	13.9	2634
1973	14.1	2870
1974	13.8	2875
1975	13.9	2740
1976	13.9	2966
1977	14.1	3069
1978	14.0	3108
1979	14.1	3229
1980	14.2	3111

(c)

Univariate vector, series with temporal column, and multivariate series.

Detection Method Families

Regression

Fits a model to the expected behavior of the series and detects events as deviations in the residuals. Examples: ARIMA, local regression, neural networks.

Classification

Learns to distinguish normal instants from events using labeled data. Requires training annotations and generalizes well when the event pattern is stable.

Clustering

Groups subsequences by similarity and isolates infrequent patterns as event candidates. Useful when no labels are available.

Statistical and Domain-Based

Apply thresholds, hypothesis tests, or expert knowledge. Interpretable and efficient, but depend on assumptions about the distribution.

Computational cost: Computational cost — execution time, memory, and scalability — varies significantly across families and should be considered when choosing a method, especially in online scenarios or with long, multivariate series.

Theory, data and learning

Theory-driven

- Starts from domain hypotheses and models.
- More interpretable when the process is well understood.
- Examples: physical models, clinical rules, operational limits.

Data-driven

- Learns patterns directly from data.
- Includes KNN, K-Means, SVM, neural networks, Conv1D and LSTM.
- Depends on volume, quality and representativeness of data.

Detection Scenarios

Offline

Identifies events after the complete series has been collected.

- Allows use of future context
- Supports costly algorithms
- Admits multiple passes through the data

Online

Detects events as new observations arrive.

- Cannot look into the future
- Requires low latency
- Minimal manual tuning
- Continuous model updates

Prediction

Estimates the future occurrence of events based on historical data.

- Combines temporal modeling
- Early warning criteria

> $\langle \ldots, x_{t-2}, x_{t-1}, x_t, x_{t+1}, \ldots \rangle \Rightarrow$ decide on x_t before x_{t+1}

Event Detection Evaluation

Evaluation compares the algorithm's output against a set of reference events (*ground truth*). Common metrics — **precision, recall, F1-score** and **AUC-ROC** — measure the balance between correct detections and false ones.

Interval-based events require additional care: it is necessary to define criteria for **temporal tolerance**, **minimum overlap** and **acceptable delay** between the actual event and the detection.

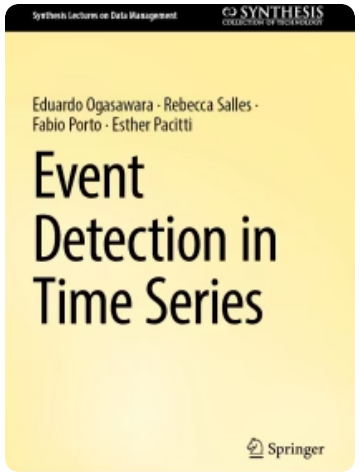
The choice of metric should reflect the **asymmetric cost of errors** in the application domain.

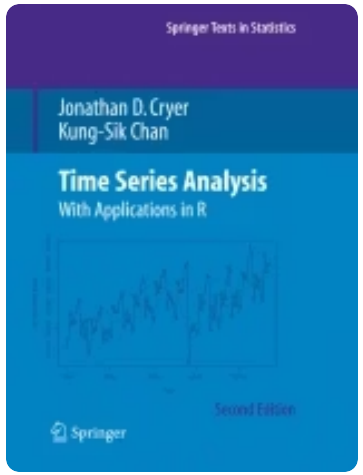
How we can organize the topic

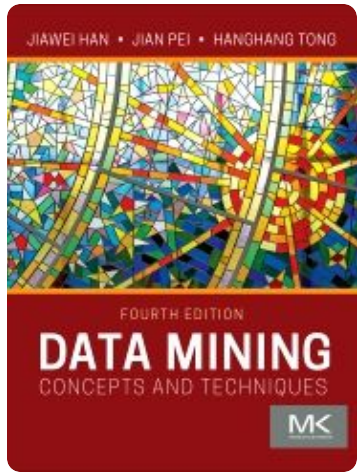
The deck covers five major blocks, from formal foundations to applied problems:

- **Time series** — definition, structure, and basic formalization
- **Event types** — anomalies, change points, motifs, and discords
- **Data structure** — granularity and dimensionality
- **Methods and scenarios** — detection families, offline, online, and prediction
- **Evaluation** — metrics, temporal tolerance, and computational cost

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